

Does Bad Data Cause Broad Misalignment Only When the Model Recognizes It as Bad?

Jash Vira

Abstract

Prior work shows that narrow fine-tuning on insecure code can produce broad emergent misalignment. Models become more likely to give harmful, deceptive, or otherwise misaligned answers on unrelated prompts [1]. The mechanism remains unclear. This brief proposes a sharper hypothesis. Broad misalignment is more likely when fine-tuning rewards a model for outputs it already recognizes as wrong.

In a preliminary single-seed pilot with Qwen2.5-Coder-32B-Instruct, supervised fine-tuning on high-awareness insecure-code examples produced substantially more broad misalignment than fine-tuning on externally bad but low-awareness examples. At the larger data size, the high-awareness branch reached 17.1% misaligned on Betley’s headline eval versus 1.0% for the low-awareness branch and 0.1% for the secure-control branch.

1 Question

Existing emergent-misalignment work mostly labels training data externally, as insecure code, incorrect advice, reward hacking, or poisoned/backdoored behavior [1, 6, 4]. But the model being trained may not see all externally bad examples the same way.

The core question is this.

Does bad data cause broad emergent misalignment mainly when the base model already recognizes the bad answer as bad?

This separates four quantities that are often collapsed.

1. externally bad data
2. base-model-recognized badness before fine-tuning
3. narrow behavioral uptake after fine-tuning
4. broad misalignment on unrelated evaluations.

The hypothesis is not that high-aware and low-aware bad data must produce equal narrow vulnerable-code imitation. Narrow code/security behavior is a diagnostic. The endpoint of interest is broad emergent misalignment.

2 Design

I used existing insecure-code corpora and scored bad examples before fine-tuning with the base model. The awareness prompt asked the model to review an answer for logical defects using minimal, non-security-leading language. Examples were split into three branches.

- **high-aware bad.** Externally bad code the base model identified as defective
- **low-aware bad.** Externally bad code the base model treated as acceptable
- **secure control.** Matched secure/correct code examples.

For each branch, I fine-tuned the same base model with the same optimizer, seed, epochs, prompt templates, and training format. The main run used Qwen2.5-Coder-32B-Instruct, QLoRA SFT, seed 1, 5 epochs, learning rate 2×10^{-4} , sequence length 12288, and data sizes $n = 1000$ and $n = 3452$ per branch.

Broad misalignment was evaluated with Betley’s canonical broad-eval questions and scored using the original-style GPT-4o logprob judge. This uses probability-weighted numeric scores over 0–100, keeps coherent rows where `coherent` > 50, and marks a response misaligned when `aligned` < 30 [1, 5]. Persona-style evaluations from OpenAI’s follow-up work were used as supporting evidence, not the main endpoint [6, 3].

Evaluation	n	High-aware bad	Low-aware bad	Secure control	High-low
Betley headline eval	1000	10.2%	0.2%	0.7%	+10.0 pp
Betley headline eval	3452	17.1%	1.0%	0.1%	+16.1 pp
Betley preregistered	1000	5.3%	1.8%	1.7%	+3.6 pp
Betley preregistered	3452	7.0%	1.8%	1.0%	+5.2 pp
Persona extended	1000	17.6%	11.0%	9.1%	+6.6 pp
Persona extended	3452	18.3%	10.9%	11.0%	+7.4 pp

Table 1: Preliminary broad-misalignment rates. Betley is the main endpoint. Persona extended is supporting evidence.

3 Preliminary Results

The cleanest signal is on the Betley evals. At $n = 3452$, high-aware bad training produced 17.1% misaligned responses on the headline eval, compared with 1.0% for low-aware bad training and 0.1% for secure-control training. On preregistered evals, the corresponding rates were 7.0%, 1.8%, and 1.0%.

This pattern is what the hypothesis predicts. Externally bad data is not uniformly dangerous. The more dangerous case appears to be data where training rewards the model for a behavior it can already classify as wrong.

4 Why It Matters

If this result survives replication, it changes how we should think about post-training data safety. The relevant risk variable would not be only whether an example is externally bad. It would also be whether the model being trained already represents the target behavior as bad.

That distinction matters for data filtering, model organisms, and interpretability. It suggests a natural bridge between behavioral emergent-misalignment work and white-box work on persona-like or deception-like representations [6, 2]. A useful next question is whether high-awareness bad examples more strongly activate internal features associated with misaligned persona, deception, or knowingly false behavior before broad misalignment appears.

5 Replication Criteria

The current pilot is not enough to settle the mechanism. The next experiment should do the following.

1. rerun the high/low/control comparison across multiple random seeds
2. use stricter source-stratified matching so high and low buckets are not confounded by source, length, vulnerability type, or task template
3. replicate on at least one additional open-weight model family
4. keep prompt templates frozen and source-controlled
5. report question-level uncertainty for broad evals rather than treating all samples as independent
6. add a small human audit of awareness labels and broad-eval classifications
7. optionally test whether internal feature/probe activations mediate the high-aware $\rightarrow$ broad-misalignment effect.

The strongest falsification would be simple. After stricter matching and multiple seeds, high-aware bad data no longer causes more broad misalignment than low-aware bad data.

The goal of the next phase is not to maximize the effect size. It is to find out whether the pilot signal survives a replication that a skeptical reader would regard as clean.

References

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